India Hydrological Outlook

Based on daily observations till 30th September 2026

Issue date: 01.10.2026

SUMMARY: In September, rainfall was mixed across the country, with very high rainfall observed in the central-east, while the north and northeast experienced lower-than-normal rainfall. Surface air temperatures were much warmer than normal across the north and south, and mixed conditions were noted in the central regions. Relative wetness showed drier-than-normal soil conditions in the north and south, and near-normal conditions elsewhere. Total runoff was high in the north, northeast, and west, and high in the central-east, whereas evapotranspiration was reduced in the north and west, and elevated in the east. For October, rainfall is predicted to be near-normal across the board, and temperatures are forecast to be warmer than normal in the north and south. Soil moisture is expected to be wetter-than-normal in the central and central-east, and total runoff is forecast to be lower-than-normal in the east and central-east.

Rainfall and Temperature:

In September, rainfall was mixed across the region, with very high rainfall observed in the central-east, while surface air temperatures were much warmer than normal across the north and south. For October, rainfall is forecast to be near-normal across the board, though the northeast is predicted to experience lower-than-normal rainfall. Temperature forecasts indicate that the central region will see warmer-than-normal conditions, while the west and south are predicted to experience much warmer-than-normal temperatures.

Soil moisture, Total runoff, and Evapotranspiration:

In September, soil moisture in the south was drier-than-normal, while the central region experienced a high runoff deficit, and the east recorded a high runoff deficit. Evapotranspiration in the west was reduced, and the central-east region showed a high runoff surplus. For October, the central region is forecast to be wetter-than-normal with a high runoff surplus, while the south and north are expected to remain drier-than-normal. Evapotranspiration across the country is forecast to be near-normal, with the central-east region showing a wetter-than-normal trend.

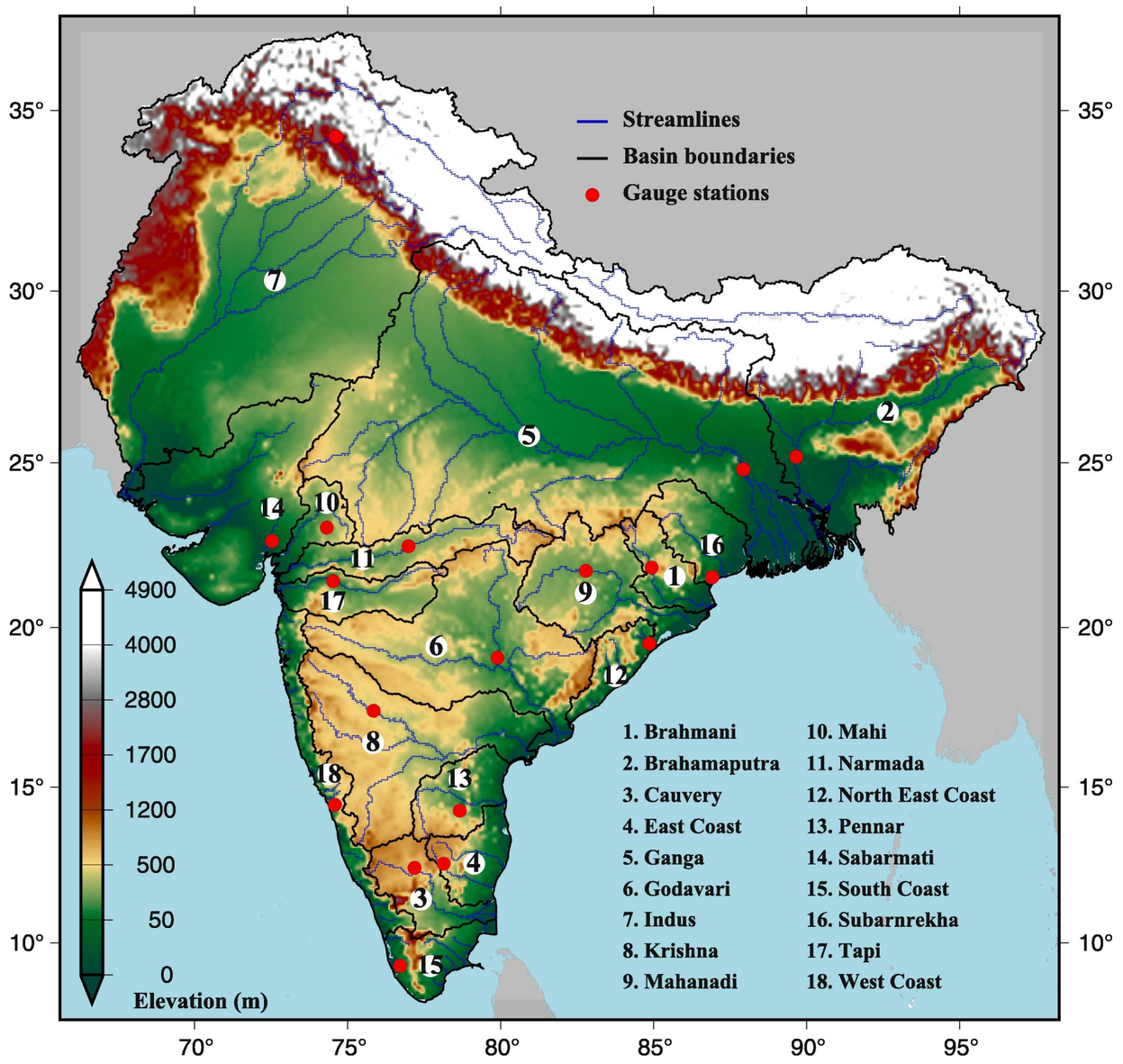


Figure. Indian sub-continental river basins. Shaded background shows elevation (m).

India Hydrological Outlook provides a comprehensive monthly summary of key meteorological and hydrological variables for current conditions alongside a four-month retrospective and one month forecast. For more information, please visit the website: www.indiahydrolook.in

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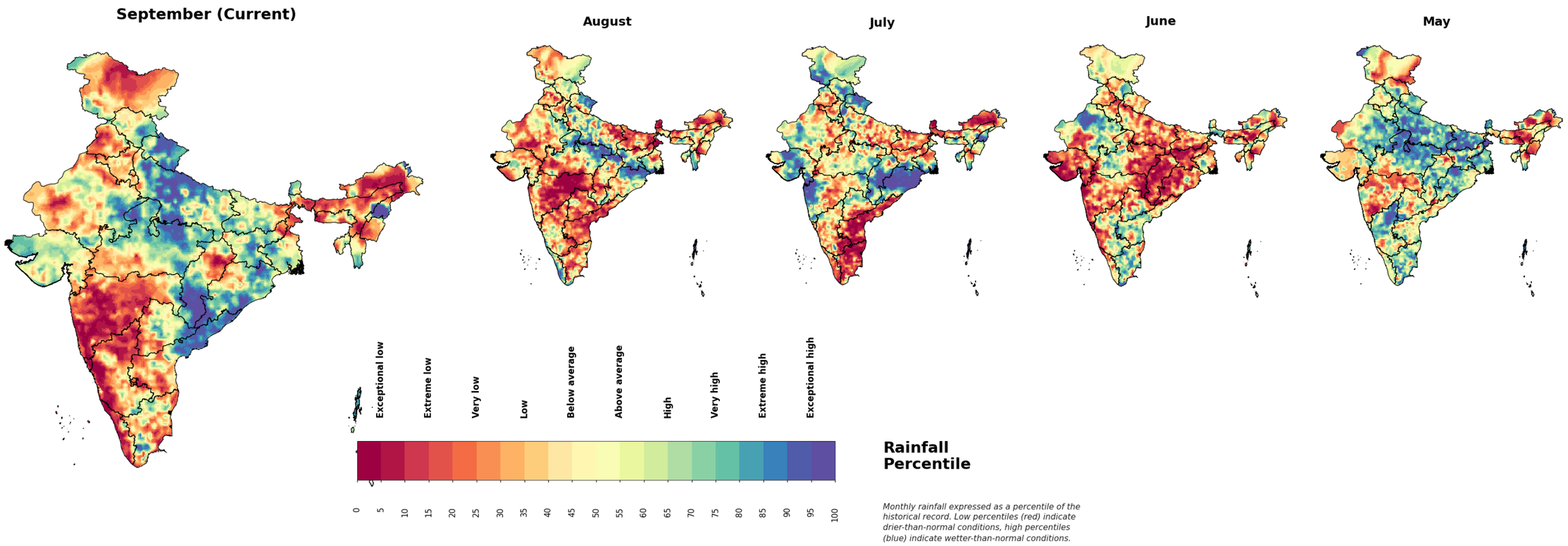
Observed and Forecast Rainfall Conditions

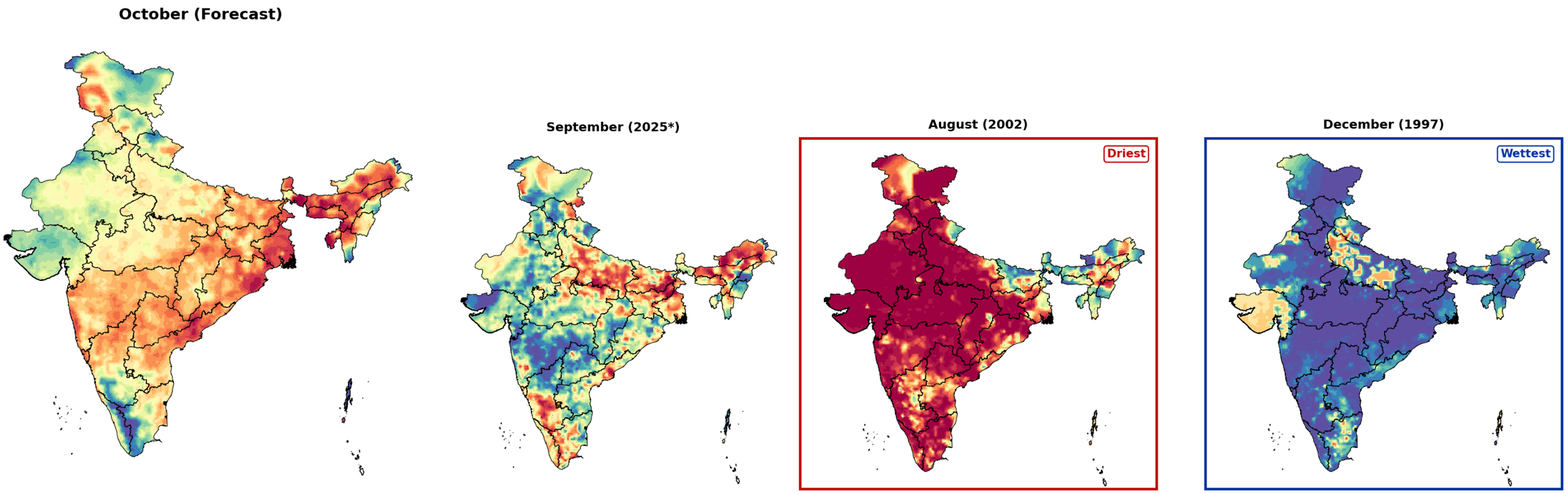
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These maps are prepared from daily gridded rainfall observations and Extended Range Forecast System (ERFS) from India Meteorological Department (IMD), expressed as total monthly rainfall in percentile for period 1955 to present. These rainfall percentile maps highlight the regions with unusual dry, wet, or near-average rainfall for that time of year. High rainfall percentiles in forecast month, along with high relative wetness (maps available on page 4) in current month, may indicate a potential region for unusual wet conditions. Areas with low precipitation percentiles in forecast month, particularly those with minimal or no rainfall along with high relative dryness, may be at risk of drought conditions.

SUMMARY: In September, rainfall showed mixed conditions across the country, with very high rainfall observed in the central-east. For October, rainfall is forecast to be lower-than-normal across the east, northeast, and south.





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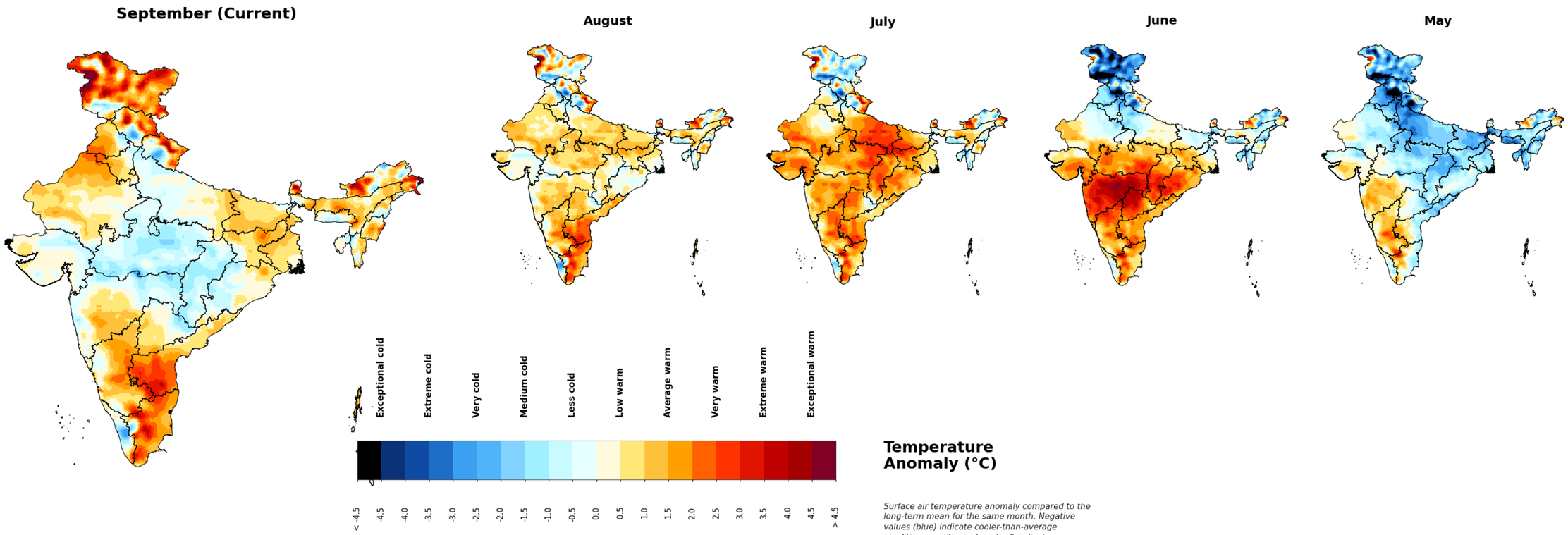
Observed and Forecast Surface Air Temperature

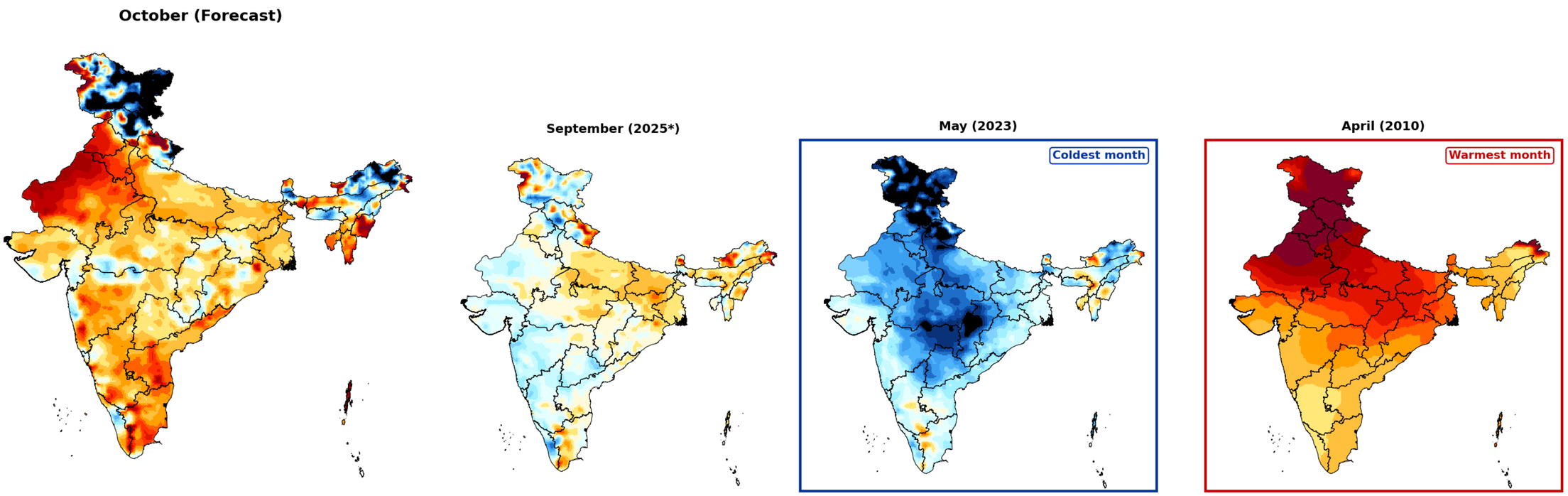
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These maps are prepared from daily gridded temperature observations and Extended Range Forecast System (ERFS) from India Meteorological Department (IMD), expressed as a monthly-averaged temperature anomaly from the historical mean (1955 to present). These temperature anomaly maps highlight regions with higher and lower temperatures than usual. Low-temperature anomalies may lead to a decrease in crop yield or delayed crop growth. Areas with higher temperature anomalies paired with relatively low rainfall and low soil moisture (maps available on page 4) may be at risk of drought conditions with increased potential evapotranspiration.

SUMMARY: Surface air temperatures were much warmer than normal across the north and south in September, while central parts experienced mixed conditions. For October, warmer-than-normal temperatures are forecast across central and east India, and much warmer-than-normal conditions are predicted in the southwest.





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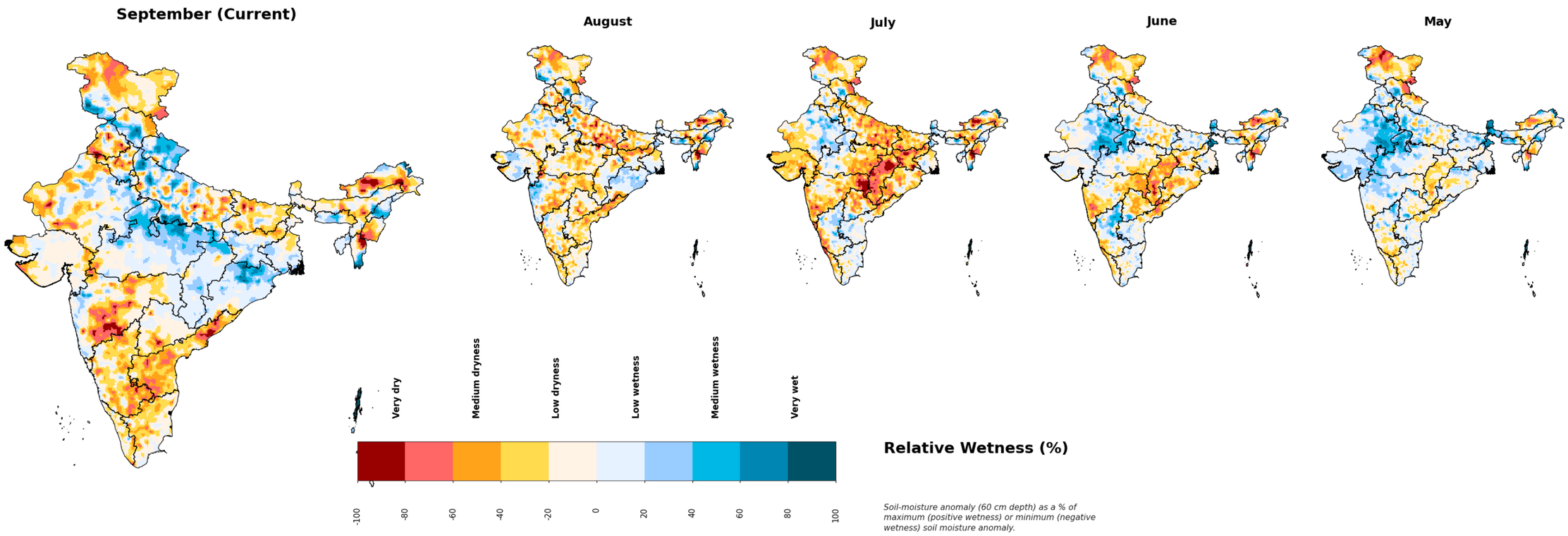
Observed and Forecast Relative Wetness

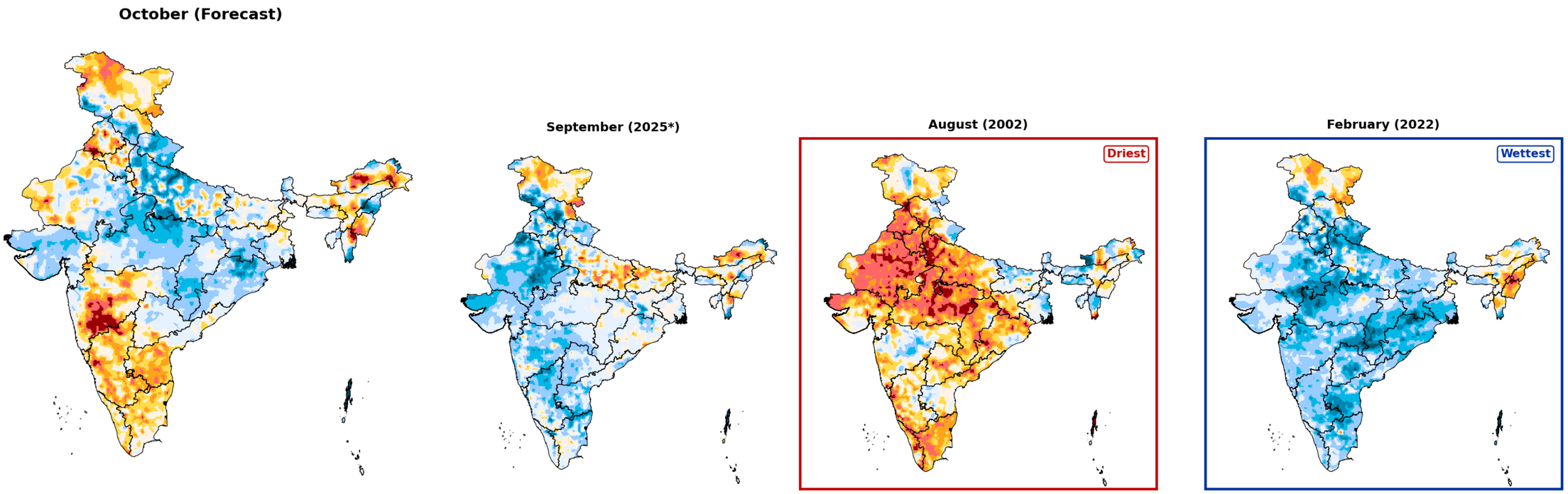
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These maps are prepared from daily simulated soil moisture (60 cm depth) using gridded meteorological forcing from observations and Extended Range Forecast System (ERFS) from India Meteorological Department (IMD), expressed as a monthly-averaged soil moisture anomaly from the historical mean (1955 to present). The soil moisture anomalies are presented as relative to historical extremes to highlight the regions with unusual wet or dry conditions. High relative wetness (wetter) in current month paired with a high rainfall percentile in forecast month may lead to above-normal flows and unusual wetter conditions in the coming days/weeks. Areas with low relative wetness (drier) in current month and no or minimal predicted rainfall in forecast month may be at risk of lower than normal flow and moisture availability.

SUMMARY: Soil moisture in September showed mixed conditions across the northern and western regions, while the central and eastern parts were near-normal. For October, the central region is forecast to be wetter-than-normal, and the central-east is predicted to be wetter-than-normal.





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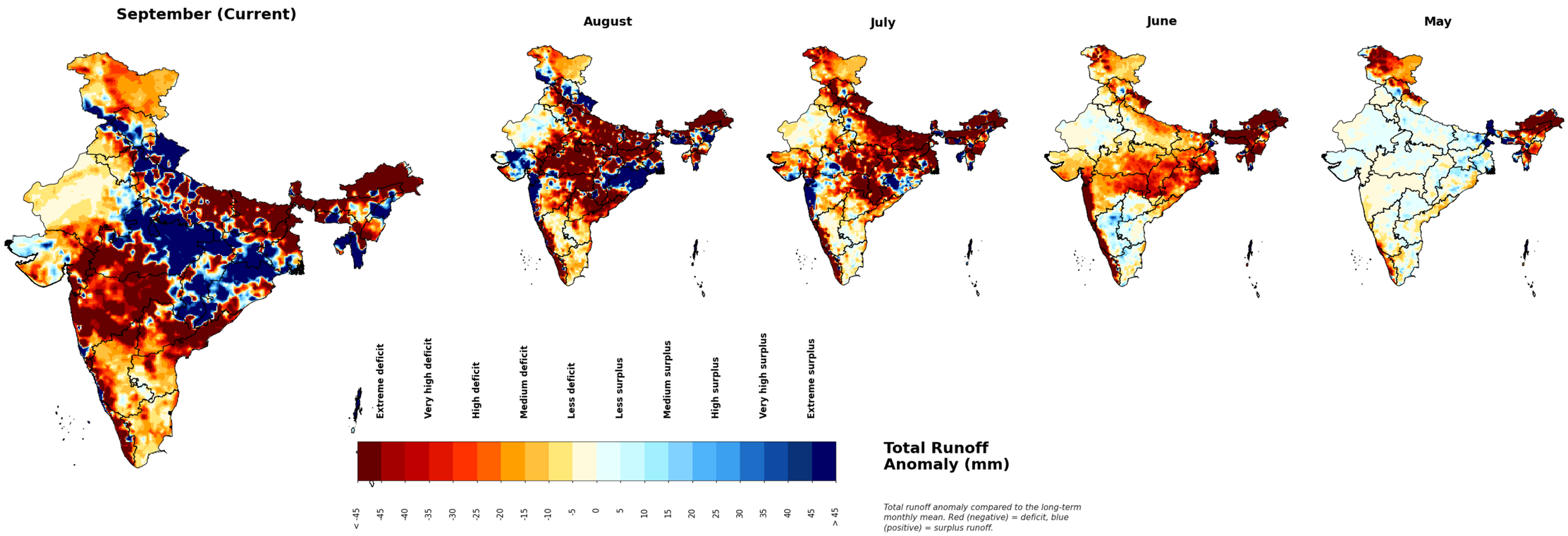
Observed and Forecast Total Runoff

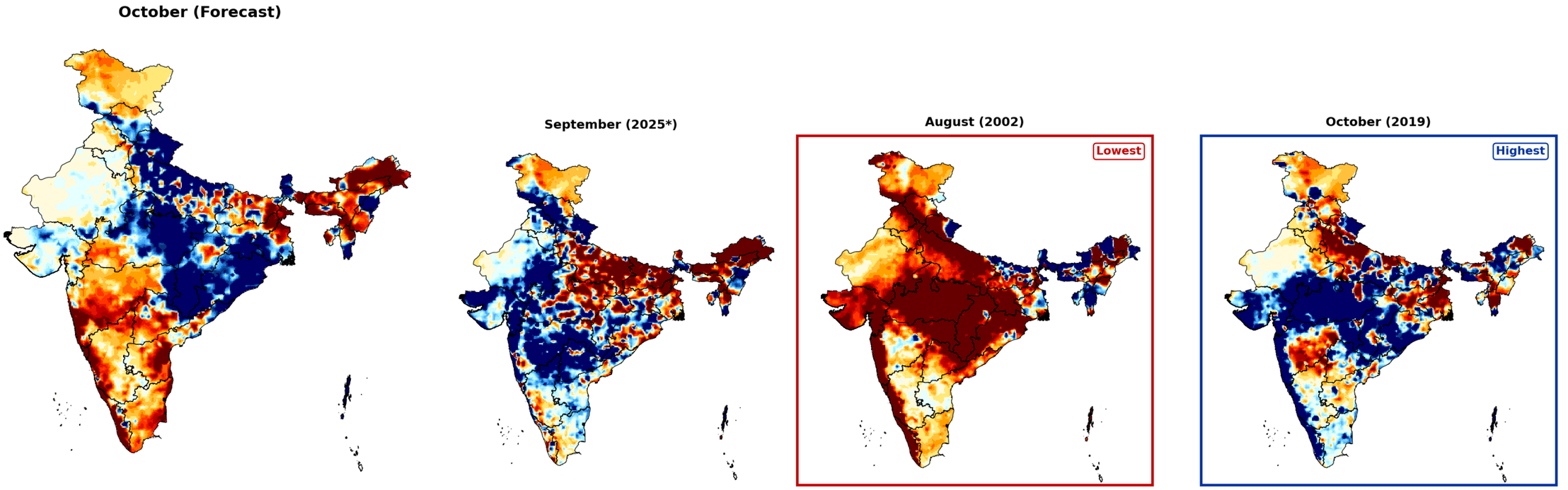
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These maps are prepared from daily simulated total runoff using gridded meteorological forcing from observations and Extended Range Forecast System (ERFS) from India Meteorological Department (IMD), expressed as a total runoff monthly anomaly from the historic mean (1955 to present). These total runoff anomalies highlight the regions with relative surplus or deficit in total runoff than typical for that time of year. A surplus in total runoff in forecast month paired with relatively higher soil moisture in current month may lead to above-normal streamflow in the coming days/weeks. Areas with a high deficit in total runoff in forecast month, particularly those with high dryness in current month, may be at risk of relatively low streamflow in the coming days/weeks.

SUMMARY: In September, a runoff deficit was observed across the north, northeast, northwest, central, east, and south. For October, the central and east regions are forecast to experience a runoff surplus.





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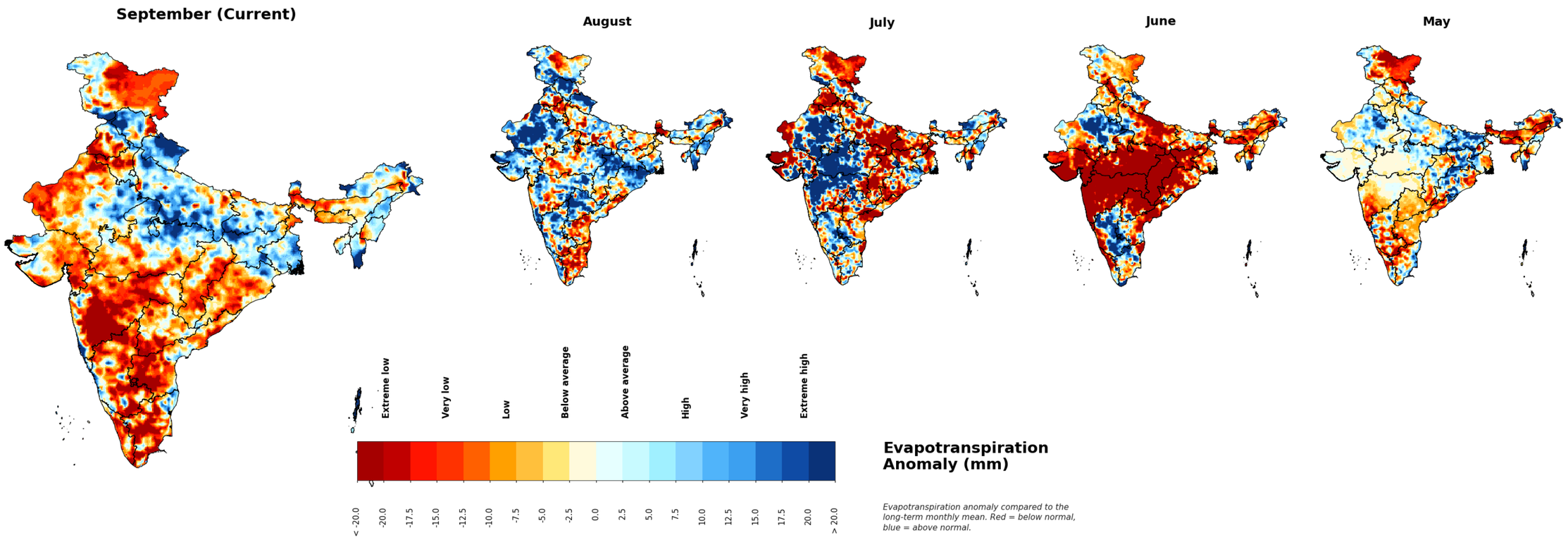
Observed and Forecast Evapotranspiration

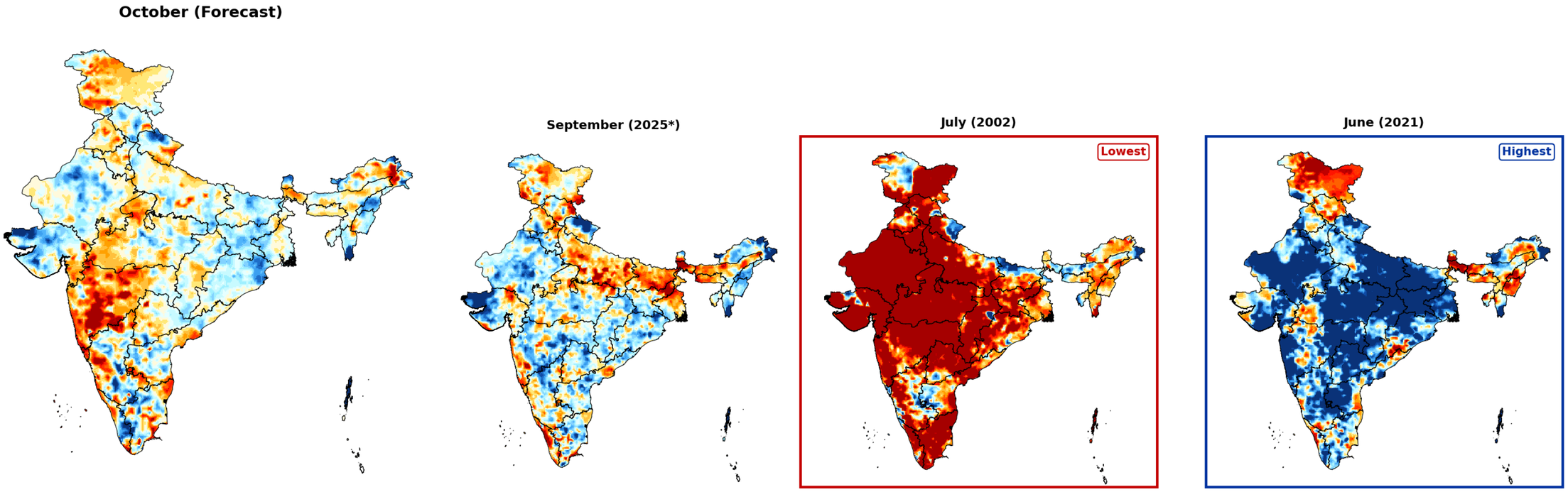
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These maps are prepared from daily simulated evapotranspiration (ET) using gridded meteorological forcing from observations and Extended Range Forecast System (ERFS) from India Meteorological Department (IMD), expressed as a total monthly ET anomaly from the historic (1955 to present). These ET anomalies highlight the regions with less or more water lost than usual from land and plants, helping to pinpoint areas at risk of extreme weather impacts, especially crop water stress. Low ET anomalies indicate increased crop water stress and relatively less moisture availability. High ET anomalies may lead to rapid depletion of soil moisture (flash drought) or may worsen the pre-existed droughts.

SUMMARY: In September, evapotranspiration conditions were mixed across the northern and central parts, while the east experienced elevated rates. For October, evapotranspiration is forecast to be near-normal across most regions.





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About the India Hydrological Outlook

This document provides a comprehensive water outlook for India, covering the current month, the past four months, and a one-month forecast. The outlook is generated using observational data and an extended-range forecasting system for meteorological variables, combined with advanced hydrological modeling tools. These tools analyze relative changes and predict water availability across various regions, offering valuable insights into hydrological situation in India. Developed by the Water and Climate Lab at IIT Gandhinagar, this outlook aims to support water resource management, planning, and decision-making by offering reliable information on rainfall, temperature, soil moisture (60 cm), runoff, and streamflow patterns across different timescales.

Datasets

The India Hydrological Outlook is made possible through the valuable cooperation of numerous data providers, whose contributions are gratefully acknowledged. Contemporary daily observations and extended range forecast data are supplied by the India Meteorological Department (IMD) https://www.imdpune.gov.in/cmpg/Griddata/Rainfall_25_Bin.html. These datasets are used for initializing hydrological models and generating reliable outlook information through statistical analysis of historical analogues. Hydrological model is calibrated and validated using daily streamflow for sufficient long duration from India-Water Information System https://indiawris.gov.in/wris#RiverMonitoring.

Hydrological Model

The India Hydrological Outlook utilizes the Variable Infiltration Capacity (VIC) model, a large-scale, semi-distributed hydrological model that computes water and energy budgets for each grid. The VIC model captures sub-grid variability in vegetation and elevation, providing a more accurate representation of hydrological processes at a regional scale. VIC model simulations are performed at a daily temporal resolution to provide accurate hydrological forecasts and enhance water resource management across India.

Disclaimer and Liability

The India Hydrological Outlook aims to ensure that all content provided is accurate and aligns with the current scientific understanding. However, the science underlying the meteorological and hydrological forecasts, as well as climate projections, is continuously evolving. As such, any forecast or prediction included in the content should not be regarded as a definitive statement of fact. To the fullest extent permitted by applicable law, the India Hydrological Outlook disclaims all warranties or representations, whether express or implied, regarding the content. Your use of the content is entirely at your own risk, and we make no guarantees that the content is error-free or suitable for your specific needs.

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India Hydrological Outlook Portal: www.indiahydrolook.in



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